Dr Francesco Sorrenti

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Personal website · ORCID

RESEARCH PROFILE

Cosmologist working on testing the foundations of the standard model (isotropy, homogeneity) with large-scale structure and Type Ia supernovae. My work combines the theoretical modelling of relativistic and velocity-field effects with its application to current data.

FIELDS OF EXPERTISE

Cosmology:	large-scale structure, galaxy clustering and magnification bias (*Euclid*), Type Ia SNe (peculiar velocities, local structure, H_0 systematics), calibrator free H_0 inference, consistency test of the cosmological model, cosmic dipoles
Data analysis:	cosmological inference, simulations and mock catalogues
Computing:	Python, JAX, Mathematica, LaTeX, C++; HPC, Git
Software:	`scoutpip` (lead developer, supernova cosmology utilities); `coffe` (contributor, full-sky galaxy number counts with lensing and RSD)

EMPLOYMENT

Oct 2025 – date	**SNSF Postdoc.Mobility Fellow** ICC, University of Barcelona, Spain
Jul – Sep 2025	**Postdoctoral Researcher** Department of Theoretical Physics, University of Geneva, Switzerland

EDUCATION

2021 – 2025	**PhD in Cosmology**, University of Geneva, Switzerland Supervisors: Profs. R. Durrer, M. Kunz. Funded by SNSF Sinergia
2018 – 2021	**MSc in Physics**, University of Perugia, Italy Final grade 110/110. Supervisor: Dr M. Gerbino
2015 – 2018	**BSc in Physics**, University of Perugia, Italy Final grade 110/110 *cum laude*. Supervisor: Prof. S. Pacetti

SELECTED PUBLICATIONS

I have authored 12 papers (9 published, 3 submitted), plus a book chapter; a selection is listed below with per-paper citation counts from INSPIRE. Papers where I am first author are marked with *. Updated lists on Google Scholar and INSPIRE. Bibliometric summary (Nov 2026; totals include collaboration papers):

- Scopus: 890 citations, h-index 6
- Web of Science: 1,075 citations, h-index 6

1. "Nonlinear velocity power spectrum: modeling the cosmological dependence on the Hubble constant and cold dark matter density", F. Lepori, R. Durrer, **F. Sorrenti** et al.
   (Submitted to OJA) (arXiv:2608.16489) *2 citations*
2. "Reconstructing the kinematics of Laniakea using Type Ia Supernovae"*, **F. Sorrenti**, E. Pastén, L. Giani
   (Submitted to JCAP) (arXiv:2607.11860) *2 citations*

3. "Exploring the cosmic microwave background dipole direction using gamma-ray bursts", O. Luongo, M. Muccino, **F. Sorrenti**
(Astronomy & Astrophysics 703, A115, 2025) (arXiv:2508.04304) *6 citations*
4. "Anchors no more: using peculiar velocities to constrain H_0 and the primordial Universe without calibrators", D. Piras, **F. Sorrenti**, R. Durrer, M. Kunz
(Journal of Cosmology and Astroparticle Physics 09, 005, 2025) (arXiv:2504.10453) *5 citations*
5. "A local infall from a cosmographic analysis of Pantheon+"*, **F. Sorrenti**, R. Durrer, M. Kunz
(Journal of Cosmology and Astroparticle Physics 12, 003, 2024) (arXiv:2407.07002) *17 citations*
6. "The low multipoles in the Pantheon+SH0ES data"*, **F. Sorrenti**, R. Durrer, M. Kunz
(Journal of Cosmology and Astroparticle Physics 04, 013, 2025) (arXiv:2403.17741) *18 citations*
7. "Euclid preparation. XL. Impact of magnification on spectroscopic galaxy clustering", Euclid Collaboration: G. Jelic-Cizmek, **F. Sorrenti**, F. Lepori, C. Bonvin, R. Durrer, M. Kunz et al.
(Astronomy & Astrophysics 685, A167, 2024) (arXiv:2311.03168) *21 citations*
8. "The Dipole of the Pantheon+SH0ES Data"*, **F. Sorrenti**, R. Durrer, M. Kunz
(Journal of Cosmology and Astroparticle Physics 11, 054, 2023) (arXiv:2212.10328) *58 citations*

RESEARCH FUNDING

2025 – 2027	**SNSF Postdoctoral Mobility Fellowship**, **CHF 103,700**
2025	**Conference funding** for CosmoFONDUE (member of the organising committee): SNSF (**CHF 2,500**) and Sacad (**CHF 10,000**)
2025	**Schmidheiny Foundation Scholarship**, for funding my participation in Cosmic Flows 2025 in Brisbane, Australia, **CHF 1,812**
2024	**ESA Scholarship**, **CHF 35,000** as team prize from "Space and Global Health" hackathon
2024	**CosmoVerse** Scholarship for visiting the University of Turin, **EUR 2,000**
2023	**Sacad** Scholarship for visiting the University of Oxford, **CHF 2,300**

ACADEMIC LEADERSHIP

***Euclid* Consortium**	**Co-lead**, SP on magnification bias extraction, DR1-KP-GC8
2021 – date	**Co-lead**, Copernican Principle project, DR1-KP-TH2
	Contributor, Dipole project, DR1-KP-TH2
	Member, DR1-KP-JC2 and TWG-WP4 (Consistency tests)
	Member, Diversity Committee (since 2026)
	Chair, then **Co-chair**, Early Career Committee (2023 – 2026)

SUPERVISION

2026	**MSc student**, University of Barcelona: project on the impact of lensing magnification on SNe Ia. Co-supervision with Prof. H. Gil-Marín

TEACHING

2023 – 2025	**Teaching Assistant**, *Mechanics II* (Bachelor), University of Geneva: 52h per year, 2 years
2022 – 2024	**Teaching Assistant**, *Theoretical Laboratory IV* (Master), University of Geneva: 52h per year, 2 years
2022	**Teaching Assistant**, *Group Theory* (Master), University of Geneva: 52h
2021 – 2022	**Teaching Assistant**, *Quantum Mechanics II* (Bachelor), University of Geneva: 1 semester

2020 – 2021	**Tutor**, *General Physics* (Engineering, 100h) and *Physics* (Environmental Sciences, 100h), Bachelor, University of Perugia

ORGANISATION OF ACADEMIC EVENTS

2025 – 2027	**CosmoFONDUE** conference series (organising committee), CERN / University of Geneva (next edition: Jan 2027)
2024	**SOC member**, *Euclid* Consortium meeting, Rome

MEMBERSHIPS & ACADEMIC SERVICE

Since 2026	**Referee**: JCAP, JHEAP
2019 – 2022	**Member**: CosmoVerse, SIF (Italian Physical Society) and AISF (Italian Association of Physics Students)

VISITS AND SCHOOLS

Mar 2025	**Visiting researcher**, University of Camerino, Italy (Cosmology Group, Prof. Luongo)
Jan 2024	**Visiting researcher**, University of Turin, Italy (Cosmology Group, Prof. Camera)
Nov – Dec 2023	**Visiting researcher**, University of Oxford, UK (Beecroft Institute, Prof. Ferreira)
Apr 2023	"Future Cosmology", CNRS thematic school, IESC Cargèse, France
Jun 2022	"Advanced *Euclid* School", Les Houches, France
Mar 2022	"Theory of Astroparticle, Cosmology and Gravitation", GGI, Italy

SELECTED TALKS

Invited talks marked with a ⋆. Full list of 22 talks here.

Nov 2026	"Fundamentals in Cosmology 2026", Cordoba, Spain
Sep 2026	"Cosmology from Home 2026", online: panelist ⋆
Jul 2026	"Exploring New Frontiers in Cosmology", GGI, Italy ⋆
Jun 2026	LUPM, Université de Montpellier, France: seminar ⋆
Mar 2025	University of Camerino, Italy: seminar ⋆
Feb 2025	"Cosmic Flows 2025", Brisbane, Australia ⋆
Jan 2025	Marseille Particle Physics Centre, France: seminar
Dec 2024	Observatório Nacional, Rio de Janeiro, Brazil (remote): seminar ⋆
Feb 2023	Cosmology Group, Tor Vergata University, Italy (remote): seminar ⋆

PUBLIC ENGAGEMENT

2023 – 2024	*Euclid* Swiss outreach
2021	POST (Science Museum, Perugia): activities for primary school children
2019 – 2021	AISF Perugia: seminars with Dr M. Gerbino (INFN) and Dr V. Vagelli (ASI), social media and website
2019 – 2021	University of Perugia: events for incoming students, theatre-based physics outreach

LANGUAGES

Italian (native); English (C1); French (B1); Spanish (B1); Catalan (A2)